

Meeting Record

Composite transformation:

More than one basic transformation is called composite transformation

2 successive Translation:

Suppose $P(x, y)$ is initial point. if we translate this point with $T_1(x_1, y_1)$ and again translate with $T_2(x_2, y_2)$, $P'(x', y')$ is the final point.

$$\begin{aligned}
 P' &= T_2(x_2, y_2) T_1(x_1, y_1) \cdot P \\
 \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} &= \begin{bmatrix} 1 & 0 & x_2 \\ 0 & 1 & y_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & x_1 \\ 0 & 1 & y_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & x_1 + x_2 \\ 0 & 1 & y_1 + y_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \\
 &= T(x_1 + x_2, y_1 + y_2) \cdot P
 \end{aligned}$$

2 successive scaling:

Suppose $P(x, y)$ is initial point. if we scale with $S(x_1, y_1)$ and again scale with $S_2(x_2, y_2)$ and $P'(x', y')$ is final point

$$P' = S \cdot P$$

$$P' = S_2(x_2, y_2) S_1(x_1, y_1) \cdot P$$

$$\begin{aligned}
 &= \begin{bmatrix} x_2 & 0 & 0 \\ 0 & y_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & 0 & 0 \\ 0 & y_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot P \\
 &= \begin{bmatrix} x_1 x_2 & 0 & 0 \\ 0 & y_1 y_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot P \\
 &= S(x_1 x_2, y_1 y_2) \cdot P
 \end{aligned}$$

2 Successive Rotation:

Suppose point $P(x, y)$ is initial point, if we rotate that point with θ_1 & θ_2 angle anticlockwise direction then $P'(x', y')$ is the final point

$$P' = R_A(\theta_2) \cdot R_A(\theta_1) \cdot P$$

$$= \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot P = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot P$$

$$P' = R(\theta_1 + \theta_2) \cdot P = \begin{bmatrix} \cos\theta_2 \cos\theta_1 - \sin\theta_2 \sin\theta_1 & -\sin\theta_2 \cos\theta_1 - \cos\theta_2 \sin\theta_1 & 0 \\ \cos\theta_1 \sin\theta_2 + \sin\theta_1 \cos\theta_2 & -\sin\theta_1 \sin\theta_2 + \cos\theta_1 \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot P$$

Translation

$$\begin{bmatrix} x' \\ y' \\ h \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \leftarrow$$

$$P' = T(t_x, t_y) \cdot P$$

Scaling

$$\begin{bmatrix} x' \\ y' \\ h \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$P' = S(S_x, S_y) \cdot P$$

Rotation:

Anticlock:

$$\begin{bmatrix} x' \\ y' \\ h \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$P' = R_A(\theta) \cdot P$$

Clockwise:

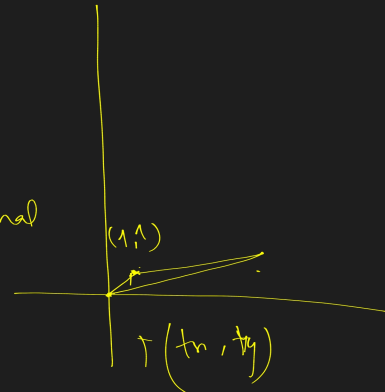
$$\begin{bmatrix} x' \\ y' \\ h \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$P' = R_c(\theta) \cdot P = R(-\theta) \cdot P$$

Q: (a) Scale the triangle with vertices A(0,0) B(1,1) C(5,2) to half of its size while keeping B(1,1) fixed.

Solⁿ: Fixed point scaling:

- (i) Translate the fixed point to origin.
- (ii) Scale the triangle half of its size
- (iii) Inverse translate the fixed point to its original position



$$\begin{aligned}
 p' &= T(1,1) \cdot S(0.5,0.5) \cdot T(-1,-1) \cdot p \\
 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_n \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_V \begin{bmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_n \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$= \begin{bmatrix} 0.5 & 1 & 3 \\ 0.5 & 1 & 1.5 \\ 1 & 1 & 1 \end{bmatrix}$$

Final coordinates of triangle

$$\underline{(0.5, 0.5) \quad (1, 1) \quad (3, 1.5)}$$

$$(10, 13)$$

$$(2, 3)$$

$$\begin{array}{l} t_n = 3 \\ t_{y_1} = 4 \end{array}$$

& then.

$$\begin{array}{l} t_{n_2} = 5 \\ t_{y_2} = 6 \end{array}$$

$$\begin{aligned} p^1 &= T(b, 10) \cdot p \\ &= \begin{pmatrix} 1 & 0 & 8 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 10 \\ 13 \\ 1 \end{pmatrix} = (10, 13) \end{aligned}$$